

Dist-Tracker: A Small Object-aware Detector and Tracker for UAV Tracking

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Abstract

*The widespread adoption of civil unmanned aerial vehicles (UAVs) has accelerated the development of anti-UAV technologies. Despite thermal infrared video enables all-weather surveillance, existing methods for multi-UAV tracking struggle with low thermal contrast, object scale variation, and erratic motion patterns. In this paper, we propose Dist-Tracker, a two-stage framework integrating a Scale-Shape-Quality (SSQ) detector based on YOLOv12 and Fusion of L2-IoU Tracker (FLIT) to address these challenges. For detection, SSQ introduces scale-aware Wasserstein distance with covariance alignment, dynamic shape-aware penalties, and adaptive gradient modulation to resolve geometric instability in small infrared targets. For tracking, FLIT synergizes IoU and L2 metrics with camera motion compensation, mitigating spatial jitter and occlusion-induced ambiguities through hybrid cost metric optimization. Comprehensive evaluations on the validation set from the Anti-UAV dataset demonstrate that our proposed framework achieves remarkable performance, with a detection AP₅₀ of 93.9% and a tracking MOTA of 77.5% in cluttered infrared environments, significantly advancing UAV swarm detecting and tracking capabilities through geometric-stable perception and motion-resilient association. Our method won **first place** in the 4-th Anti-UAV challenge Track3 (**tracking MOTA:81.32%** on the official test set).*

1. Introduction

In recent years, civil unmanned aerial vehicles (UAVs) [35], commonly called drones, have experienced rapid development and have been widely applied in various fields, including agriculture, environmental monitoring, disaster management, logistics, and even entertainment [18]. However, the proliferation of UAVs has also raised significant security concerns, as misuse or unauthorized operations pose risks ranging from privacy violations to physical infrastructure damage. To mitigate these threats, the development of anti-UAV technologies [10, 11] has become imperative. Among these, multi-UAV tracking stands out as a critical strategy for enabling continuous surveillance of drone swarms through trajectory analysis in thermal infrared video sequences. This capability is essential for preemptive threat detection and coordinated countermeasures in multi-drone scenarios.

The core objective of multi-UAV tracking is to continuously detect and track multiple UAVs across video frames, initialized by known detections in the first frame. While sharing similarities with general multi-object tracking (MOT) [17], multi-UAV tracking presents unique challenges specific to thermal infrared imaging and UAV dynamics: (1) UAVs frequently exit the camera's field of view or suffer occlusions due to environmental obstructions. (2) In thermal imagery, UAVs exhibit minimal texture contrast with the background, severely limiting the reliability of visual features for detection. (3) Most commercial drones share nearly identical thermal signatures and occupy few pixels, complicating instance discrimination. (4) UAVs can execute abrupt accelerations, hovering, or non-linear flight paths that defy conventional motion models. These compounded challenges make robust multi-UAV tracking in thermal infrared video still a significant challenge. To this end, in this paper, we propose a novel strategy, Dist-Tracker,

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which involves a detection model and a tracking model tailored to multiple UAV tracking.

In infrared small target detection, challenges closely related to UAV tracking persist. Some data analysis reveals inherent limitations in detecting small UAVs: low thermal contrast between uniform drone signatures and background texture, regression instability amplified by minimal pixel occupancy, and severe detection failures with high false-negative rates and false-positive alarms in scenarios involving complex background clutter, motion blur, and thermal crossover [23]. While YOLOv12’s area attention mechanism partially suppresses noise, the default CIoU loss exhibits critical flaws [24]: (1) the joint gradient updates of center distance and aspect ratio, combined with rigid bounding box constraints, cause unsuitable IoU calculations and shape modeling failures during abrupt motions; (2) the equal weight allocation of IoU losses on targets of different sizes will lead to the scale perception bias of targets [32]. For this, we propose the Scale-Shape-Quality (SSQ) loss to address these bottlenecks through multiple geometric modeling: (1) scale-aware Wasserstein distance to address positional sensitivity in UAV detection; (2) covariance matrices to model scale uncertainty; (3) Shape-Aware loss matching the target shape of infrared drones better; (4) adaptive gradient modulation based on anchor box quality assessment. On the Anti-UAV dataset, SSQ significantly reduces missed detections for challenging targets while maintaining robust localization fidelity.

In tracking, data association is a critical component. While previous work [1, 3, 33] has predominantly used Intersection over Union (IoU) for detection-trajectory association, we found it to be suboptimal for UAV scenarios. Through dataset analysis, we observe two inherent conflicts: First, the small target size in UAV imagery causes positional jitter that destabilizes IoU measurements. Second, frequent occlusion and scale changes lead to matching ambiguities. These factors collectively result in significantly higher ID switches compared to ground-based MOT when using standard IoU association. Motivated by this, we propose FLIT (Fusion of L2-IoU Tracker), a hybrid distance metric that synergistically combines target geometry and spatial coherence. The core innovation lies in augmenting the IoU metric with L2 distance between predicted target positions and detected bounding boxes. This dual-component design specifically addresses the observed limitations: the IoU component maintains geometric consistency, while the L2 term compensates for positional uncertainties caused by small targets and abrupt motions.

In summary, our main contributions are as follows:

- We propose a novel loss function, specifically designed for UAV data, called SSQ loss. This approach effectively enhances detection accuracy through multiple geometric modeling and significantly reduces missed detections.

- We design an innovative tracking method named FLIT. In this tracking algorithm, a hybrid distance metric is introduced for data association, combining L2 distance and IoU, which effectively balances target overlap and positional uncertainty.
- Experimental results demonstrate the effectiveness of the proposed Dist-Tracker. It significantly improves the performance of multi-UAV tracking and achieves **the first place** in the 4-th Anti-UAV Challenge. Extensive ablation studies further validate the effectiveness and superiority of our method.

2. Related Work

2.1. Object Detection

The rapid evolution of real-time object detection frameworks has been driven by the YOLO series. YOLOv1 [21] pioneered grid-based regression for unified localization and classification, achieving 45 FPS but struggling with small targets. Subsequent iterations introduced critical improvements: YOLOv2 [19] integrated anchor boxes and batch normalization, YOLOv3 [20] adopted Darknet-53 with FPN for multi-scale prediction, and YOLOv4 [4] enhanced robustness via CSPNet and Mosaic augmentation. Recent versions [15, 26–28] like YOLOv7 introduced programmable gradient information to address gradient degradation. Most recently, YOLOv12 [24] integrated area attention and residual efficient layer aggregation network (RELAN) to construct the first attention-centric YOLO framework. Combined with FlashAttention [6] for memory efficiency, lightweight design adjustments (*e.g.*, removing positional encoding and adjusting MLP ratios), this framework achieves superior performance over traditional CNN architectures while maintaining real-time inference capabilities.

The advancement of infrared small-target detection has paralleled the evolution of YOLO frameworks, driven by the unique challenges of low signal-to-noise ratios (SNR), thermal noise interference, and sub-pixel target scales in UAV scenarios. Early approaches focused on frequency-domain enhancement, such as wavelet-based cross-scale fusion modules (WCFM) [8] to isolate targets from cluttered backgrounds by decomposing images into multi-frequency subbands and adaptively amplifying high-frequency target details. Subsequent breakthroughs integrated attention mechanisms for context-aware detection. DNANet [14] leveraged hierarchical attention layers to amplify subtle thermal signatures, while UIU-Net [30] introduced an interactive-cross attention module to fuse multi-scale features, effectively enhancing global-local contrast for morphologically unstable targets. Recent YOLO-based frameworks like YOLO-MST [32] introduced channel-detail attention, dynamically reweighting spectral and spatial fea-

tures to suppress false alarms from cloud edges and thermal reflections.

Regression loss in object detection aims to localize the target. While typical losses like GIoU [22] and CIoU [36] have demonstrated effectiveness for general cases, their performance degrades significantly when handling small targets. Notably, DIOU [22] addresses partial limitations by incorporating centroid distance measurement alongside overlap area, yet still struggles with aspect ratio sensitivity. To address this limitation, Shape-IoU [31] proposes a geometric shape-aware loss that explicitly models bounding box orientation and aspect ratio via covariance matrices, enhancing robustness against irregular shapes. Normalized wasserstein distance (NWD) [29] extends this concept by measuring shape similarity through Gaussian distribution alignment, enabling scale-invariant regression. Meanwhile, Wise-IoU [25] dynamically prioritizes localization errors for hard samples (*e.g.*, occluded or low-contrast UAVs) via adaptive error weighting based on gradient gain modulation.

2.2. Object Tracking

With the rapid advancement of object detection models, tracking-by-detection has become one of the dominant approaches for MOT due to its efficient performance and flexible modular architecture. Tracking-by-detection methods first use object detectors to obtain bounding boxes and other object-related information, then associate those detections across consecutive frames based on appearance or spatial information to construct complete trajectories. SORT [3] adopts a simple and effective design relying on spatial information and motion prediction for data association. Specifically, SORT uses a Kalman filter motion model [12] to predict the object positions in the current frame based on their previous locations and motion states. Then, the Hungarian algorithm [13] is employed to associate these predicted positions with detections in the current frame based on IoU, thus forming continuous object trajectories. ByteTrack [33] improves on SORT by proposing a two-stage association method that emphasizes the importance of low-confidence detections. Specifically, it first matches high-confidence detections to reduce incorrect associations, and subsequently matches low-confidence detections to recover targets missed in the first step (*e.g.*, occluded targets). This strategy effectively reduces ID switches and enhances tracking performance. BoT-SORT [1], built upon ByteTrack, further introduces a camera motion compensation (CMC) module to address camera movement and dynamic backgrounds. In addition, it refines the Kalman filter parameters to correct prediction deviations and integrates appearance information into the association process, significantly improving tracking accuracy in dynamic scenarios.

Considering the small size and highly similar appearance of UAVs in infrared images, we present FLIT (Fusion of

L2-IoU Tracker), an enhanced tracking algorithm building upon the SORT. FLIT combines IoU and L2 distance metrics in the association process, effectively addressing challenges such as small target size and occlusions without relying on appearance features.

3. Proposed Method

In the following, we introduce in detail our proposed infrared small target detection and tracking framework, named Dist-Tracker. The overall architecture is illustrated in Figure 1.

3.1. Object Detection

We select the state-of-the-art YOLOv12 as the baseline architecture due to its intrinsic adaptability to infrared small target detection, where the Area Attention (A²) mechanism demonstrates superior capability in capturing low-contrast thermal signatures [24]. However, its default CIoU [36] localization loss exhibits critical limitations. To address this, we formulate an optimization loss function to tackle these bottlenecks.

3.1.1. Review of the loss in YOLOv12

YOLOv12 employs CIoU loss L_{CIoU} [36] for localization, Focal Loss [16] L_{Focal} for classification, and objectness loss L_{obj} for objective confidence:

$$L_{det} = 7.5L_{CIoU} + 0.5L_{Focal} + 1.5L_{obj}. \quad (1)$$

The L_{CIoU} can be formulated as:

$$L_{CIoU} = 1 - \text{IoU} + \frac{\rho^2(b, b^{gt})}{c^2} + \alpha v, \quad (2)$$

where ρ denotes Euclidean distance between the center points of the predicted box b and ground-truth box b^{gt} . c denotes the diagonal length of the minimum enclosing box. v denotes the aspect ratio penalty and α denotes its weight.

However, as mentioned earlier, the CIoU loss and its improved versions still have limitations in the infrared UAV object detection task. In the geometric penalty terms of enhanced IoU variants, the center distance penalty and aspect ratio penalty generate competitive gradient signals during backpropagation, causing periodic oscillations in parameter updates. This deficiency in geometric decoupling significantly degrades regression stability in regions with abrupt scale variations [34, 36].

3.1.2. Scale-Shape-Quality Loss

Considering the challenges in infrared UAV object detection, such as edge blurring and background clutter between UAVs due to low thermal signal-to-noise ratio, difficulty in shape modeling, and extreme scale sensitivity caused by a

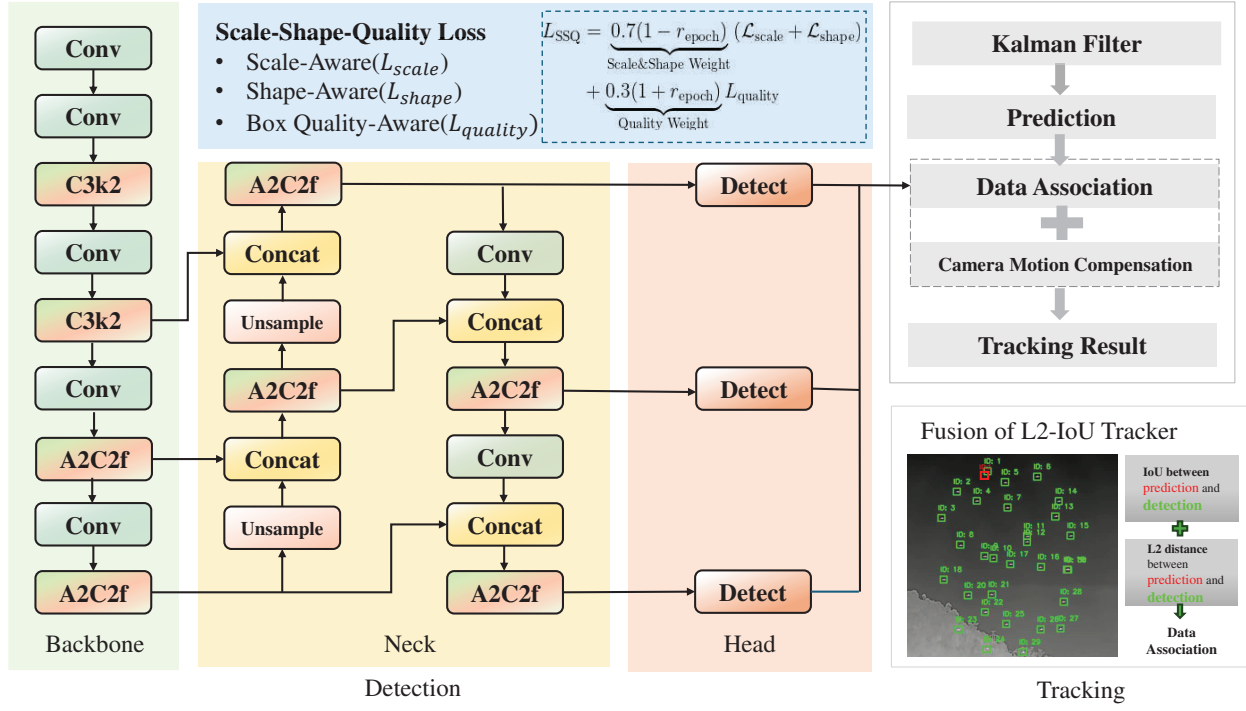


Figure 1. The overall architecture of our proposed model. The left part illustrates the detection module, while the right part shows the tracking module. In the detection part, we adopt the main framework of YOLOv12 and redesign the loss function L_{SSQ} specifically for UAV data. In the tracking part, we employ a data association algorithm that integrates IoU and L2 distance. In the figure, green indicates detection results, and red represents predictions.

small number of pixels occupied by small targets. We propose the SSQ loss to alleviate these issues through multi-modal geometric modeling with three core improvements based on the CIoU loss.

Scale-Aware. In order to solve the challenges of small target and target scale variation in UAV detection, we introduce a scale-aware Wasserstein distance following [9, 29]:

$$L_{\text{scale}} = \underbrace{\|\mu_p - \mu_t\|_2^2}_{\text{Center Alignment Term}} + w_s \cdot \underbrace{\left\| \Sigma_p^{1/2} - \Sigma_t^{1/2} \right\|_F^2}_{\text{Scale Consistency Term}}, \quad (3)$$

where $\mu_p = (x_p, y_p)$ and $\mu_t = (x_t, y_t)$ denote coordinates of the predicted center and the ground truth center, respectively. Covariance matrix $\Sigma_p = \text{diag}(w_p^2, h_p^2)$ encodes the confidence of the model for the predicted bounding box scale, while $\Sigma_t = \text{diag}(w_t^2, h_t^2)$ reflects the inherent distribution characteristics of the ground truth scale. *Center Alignment Term* addresses positional sensitivity in UAV detection and mitigates boundary discontinuity issues by providing gradient-smooth supervision for center regression; while *Scale Consistency Term* is Frobenius norm difference, modeling scale uncertainty through covariance matrices. This term enables linear response to scale changes, resolving the nonlinearity of IoU in

small-scale regimes. w_s denotes the weight of the *Scale Consistency Term*. Empirical studies show that $w_s > 0.5$ emphasizes scale matching critical for small targets.

Shape-Aware. While Scale-Aware loss addresses scale issue, aspect ratio modeling is also crucial due to the unique geometric patterns of UAV targets. For instance, our statistical analysis reveals that UAV instances in the training set predominantly maintain an aspect ratio of about 1.2:1. Motivated by this, we introduce a Shape-Aware loss [31]:

$$L_{\text{shape}} = \underbrace{\frac{2w_t^{\text{scale}}}{w_t^{\text{scale}} + h_t^{\text{scale}}} \left(\frac{|w_p - w_t|}{\max(w_p, w_t) + \epsilon} \right)^2}_{\text{Width Term}} + \underbrace{\frac{2h_t^{\text{scale}}}{w_t^{\text{scale}} + h_t^{\text{scale}}} \left(\frac{|h_p - h_t|}{\max(h_p, h_t) + \epsilon} \right)^2}_{\text{Height Term}} \quad (4)$$

where *scale* controls aspect-ratio sensitivity, amplifying width/height error penalties for elongated targets [31]. The fixed *scale* amplifies aspect ratio differences, better matching the target shape of infrared drones and the $\max(\cdot)$ operation dynamically adjusts penalty intensity per sample. ϵ ensures numerical stability for sub-pixel targets, critical for

small UAVs. On the macro level, the *Width Term* adaptively emphasizes width errors for wide targets ($w_t/h_t > 1$) and the *Height Term* emphasizes height errors for tall ones ($w_t/h_t < 1$), mimicking human visual perception of shape deviation [31].

“Box Quality”-Aware. Traditional IoU variants often model gradient contributions based on localization quality, leading to more focus on hard samples. Our improved Quality loss introduces adaptive gradient modulation based on anchor quality assessment. The formulation aligns with the core principles of dynamic focusing in [25], while enhancing numerical stability for UAV detection:

$$L_{quality} = \underbrace{\frac{\beta}{\sigma^{\beta-1+\epsilon}}}_{\text{Gradient Focus Factor}} \cdot \underbrace{(1 - \text{IoU})^\sigma}_{\text{Base IoU Loss}} \quad (5)$$

where σ (focus parameter) controls the gradient decay rate for low-IoU targets. $\mathbb{E}[1 - \text{IoU}]$ is the sliding mean of IoU errors over a 100-iteration window, normalizing β to batch statistics. β represents “Outlier Metric”, using the dynamic gradient modulation mechanism prioritizes semi-hard samples, which can be formulated as:

$$\beta = \frac{1 - \text{IoU}}{\mathbb{E}[1 - \text{IoU}] + \epsilon}. \quad (6)$$

Here, ($\beta > 1$) suppresses gradients for hard samples (e.g., occluded UAVs) to mitigate overfitting, ($\beta < 1$) preserves baseline supervision for easy samples, and ($\beta \approx 1$) maximizes gradient focus on critical median-quality anchors [25].

Multiple Loss Combination. Finally, we devise a phased training strategy that progressively shifts focus from geometric stability to localization precision. The SSQ loss can be formulated as:

$$L_{SSQ} = \underbrace{0.7(1 - r_{\text{epoch}})}_{\text{Scale\&Shape Weight}} (\mathcal{L}_{\text{scale}} + \mathcal{L}_{\text{shape}}) + \underbrace{0.3(1 + r_{\text{epoch}})}_{\text{Quality Weight}} L_{\text{quality}} \quad (7)$$

$$r_{\text{epoch}} = \frac{\text{current_epoch}}{\text{total_epochs}} \quad (8)$$

By using the epoch-dependent term r_{epoch} , the $L_{\text{scale}} \& L_{\text{shape}}$ can establish stable geometric constraints in early training phases, and gradually transition to high-quality positioning with L_{quality} [29]. Building upon YOLOv12’s multi-task architecture, we reformulate the detection loss as:

$$L_{\text{det}} = \lambda_{\text{ssq}} L_{\text{SSQ}} + \lambda_{\text{cls}} L_{\text{Focal}} + \lambda_{\text{obj}} L_{\text{obj}} \quad (9)$$

Integrated with YOLOv12, SSQ Loss simultaneously addresses the scale sensitivity and target aspect ratio of UAV through multiple geometric modeling. We retained YOLOv12’s original 7.5:0.5:1.5 fixed weight ratio [7, 24].

Algorithm 1 Pseudo-code of FLIT

Input: A video sequence V ; object detector Det ; fusion cost matrix C_{final} ; confidence matrix $\mathcal{S}_{\text{conf}}$; confidence score threshold τ ; new track score threshold η

Output: Tracks $\mathcal{T}$ of the video

```

1: Initialization:  $\mathcal{T} \leftarrow \emptyset$ 
2: for frame  $f_k$  in  $V$  do
3:   /* Handle new detections */
4:    $\mathcal{D}_k \leftarrow Det(f_k)$ 
5:    $\mathcal{D}_{\text{conf}} \leftarrow \emptyset$ 
6:   for  $d$  in  $\mathcal{D}_k$  do
7:     if  $d.\text{score} > \tau$  then
8:       /* Store detections over threshold */
9:        $\mathcal{D}_{\text{conf}} \leftarrow \mathcal{D}_{\text{conf}} \cup \{d\}$ 
10:    end if
11:  end for
12:  /* Predict new locations of tracks */
13:  for  $t$  in  $\mathcal{T}$  do
14:     $t \leftarrow KalmanFilter(t)$ 
15:     $t \leftarrow MotionCompensation(t)$ 
16:  end for
17:  /* data association */
18:   $C_{\text{iou}} \leftarrow IOUDist(\mathcal{T}, \mathcal{D}_{\text{conf}})$ 
19:   $C_{\text{l2}} \leftarrow L2Dist(\mathcal{T}, \mathcal{D}_{\text{conf}})$ 
20:   $C_{\text{iou-l2}} \leftarrow Fusion(C_{\text{iou}}, C_{\text{l2}})$  ▷ Eq. 10
21:  /* Calculating Fusion Cost */
22:   $C_{\text{final}} \leftarrow Fusion(C_{\text{iou-l2}}, \mathcal{S}_{\text{conf}})$  ▷ Eq. 11
23:  Linear assignment by Hungarian’s alg. with  $C_{\text{final}}$ 
24:   $\mathcal{D}_{\text{remain}} \leftarrow$  remaining object boxes from  $\mathcal{D}_{\text{conf}}$ 
25:   $\mathcal{T}_{\text{remain}} \leftarrow$  remaining tracks from  $\mathcal{T}$ 
26:  /* Delete unmatched tracks */
27:   $\mathcal{T} \leftarrow \mathcal{T} \setminus \mathcal{T}_{\text{remain}}$ 
28:  /* Initialize new tracks */
29:  for  $d$  in  $\mathcal{D}_{\text{remain}}$  do
30:    if  $d.\text{score} > \eta$  then
31:       $\mathcal{T} \leftarrow \mathcal{T} \cup \{d\}$ 
32:    end if
33:  end for
34:  Return  $\mathcal{T}$ 
35: end for

```

3.2. Object Tracking

The classic SORT [3] algorithm is generally simple and effective in multi-object tracking tasks but shows limitations when tracking drones. On the one hand, relying solely on the IoU metric for data association becomes inadequate, particularly when the targets are moving rapidly or are oc-

cluded, thereby increasing the likelihood of incorrect associations and target loss. On the other hand, the motion of the drone causes frequent changes in the camera’s perspective, which further exacerbates the problem of frequent ID switches. Therefore, we propose a novel tracker, FLIT, which builds upon SORT [3] and introduces several improvements to address the unique challenges of drone videos. Specifically, we introduce a data association strategy that combines IoU and L2 distance metrics with a low-threshold design, and incorporate a camera motion compensation (CMC) mechanism. For the motion prediction model, we adopt the improved Kalman filter model from BoT-SORT [1].

IoU - L2 Fusion. We develop a matching method that integrates both spatial proximity and shape overlap between objects. In the Anti-UAV datasets, objects typically appear small, so even slight positional deviations can cause a rapid decline in IoU. Additionally, drones move quickly, limiting the accuracy of Kalman filter predictions, further increasing positional discrepancies and causing IoU values to drop sharply, leading to matching failures. To address these issues, we introduce a combined matching metric integrating the IoU distance and the L2 distance between object centers during data association.

$$C_{IoU-L2} = \gamma C_{L2} + (1 - \gamma) C_{IoU} \quad (10)$$

where C_{IoU-L2} represents the combined cost Matrix, C_{IoU} is the IoU distance measuring the overlap between the predicted tracklets bounding boxes and detected ones, C_{L2} is the normalized L2 distance measuring the Euclidean distance between the centers of the predicted and detection boxes, and γ is the weighting factor. Based on multiple experiments, we found that setting the value of γ to 0.25 yields the best overall performance in terms of tracking accuracy and stability.

Following the method in BoT-SORT [1], the obtained cost matrix C_{IoU-L2} is further refined by incorporating the detection confidence.

$$C_{final} = 1 - (1 - C_{IoU-L2}) S_{conf} \quad (11)$$

where C_{final} is the final cost matrix, S_{conf} is the confidence matrix expanded to match the shape of C_{final} .

The underlying rationale behind IoU - L2 Fusion approach is that objects typically move limited distances between consecutive frames. Even when IoU values decline sharply due to minor positional errors, the L2 distance between object centers remains relatively small. Therefore, incorporating L2 distances into the matching metric mitigates the issue of rapidly decreasing IoU caused by small positional deviations, to some extent reducing ID switches and enhancing the continuity and stability of UAV tracking.

Camera Motion Compensation. Considering that Anti-UAV dataset consists of infrared videos, frequent cam-

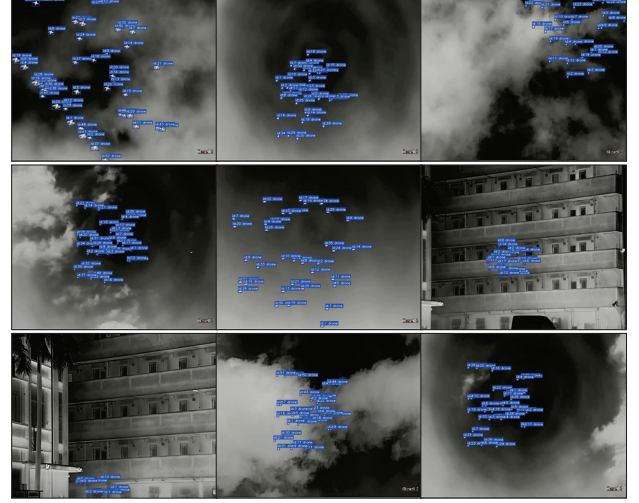


Figure 2. UAV tracking results across different scenarios

era movements cause substantial variations in target positions, significantly impacting tracking stability. We introduce a global motion compensation (GMC) method based on Sparse Optical Flow, specifically designed to correct positional deviations in Kalman Filter predicted bounding boxes caused by camera motion. Specifically, we directly utilize the Sparse Optical Flow-based motion compensation methods provided by the OpenCV Library [5]. First, keypoints between consecutive frames are detected, and their sparse optical flow is computed to capture global motion patterns. Then, the affine transformation matrix representing the camera’s global movement between frames is estimated via affine transformation estimation function. This affine transformation is applied to when predicting the bounding boxes, correcting positional deviations induced by camera motion. This compensation strategy improves the accuracy and stability of trajectory predictions in infrared drone videos, effectively reducing data association failures and tracking errors caused by frequent camera motion.

4. Experiments

4.1. Experimental Setup

In this paper, we tested our proposed Anti-UAV detection and tracking solution using two NVIDIA GeForce RTX 4090 GPUs.

Dataset. We utilized the training subset of the 4th Anti-UAV dataset, which consists of 200 video sequences. The dataset includes video sequences with various challenging scenarios, such as complex dynamic backgrounds, small target sizes, target disappearance and reappearance, target occlusion, and the emergence of new targets. The maximum number of targets per frame can exceed 40. We randomly

split the dataset into a training set with 160 video sequences and a validation set with 40 video sequences. Unless otherwise specified, all the following results are obtained by training on the training set and testing on the validation set. **Data Augmentation.** Tailored for infrared small target detection and UAV dynamics, some data augmentation strategies are employed. We used color space enhancements via HSV adjustments ($H = 0.015, S = 0.7, V = 0.4$), which improves thermal discriminability without color distortion. Geometric transformations including translation ($translate = 0.1$) is applied for abrupt motion simulation, and scaling ($scale = 0.5$) is used for covering targets of different scales. For dense multi-drone scenarios, Mosaic is used to mitigate occlusion and complex background issues. **Metric.** Following the evaluation rules of 4-th Anti-UAV Workshop & Challenge, we use average precision (AP) to evaluate the detection ability of the model and multiple object tracking accuracy (MOTA) to evaluate the tracking ability. The MOTA [2] is defined as:

$$MOTA = 1 - \frac{FP + FN + IDS}{GT} \quad (12)$$

where FP is the number of false positives (detected but non-existent targets), FN is the number of false negatives (missed targets), IDS is the number of ID switches (assigning a new ID to an already tracked target), and GT is the total number of ground truth objects.

4.2. Main Results

Table 1 presents the main results of Track-3 of the 4-th Anti-UAV competition. As shown in the table, our method (earth-insights) achieved the highest MOTA score, outperforming other competing methods. Specifically, we integrated the SSQ loss into our best detection model and trained it with an image size of 1280 pixels, subsequently combining it with the FLIT tracker. This approach yielded favorable results on the test set. Figure 2 illustrates our tracking performance across various scenarios, demonstrating the robustness and accuracy of our proposed method in diverse scenarios.

4.3. Ablation study

4.3.1. Object Detection

The ablation experiments demonstrate the effectiveness of different detection improvements over the baseline model. The experimental results are presented in Table 2. The baseline (YOLOv12) achieves 84.5% AP_{50} and 46.6% AP_{50-95} at an input size of 640. Introducing Shape-IoU and NWD individually improves AP_{50} to 86.1% and 86.6%, respectively, though NWD slightly reduces AP_{50-95} compared to Shape-IoU. Combining both modules (“+Shape-NWD”) yields suboptimal results, suggesting potential incompatibility between these two strategies. Notably, our proposed SSQ

Table 1. Main results of Track3 with 100 videos

Rank	Team Name	Score(MOTA)
1	earth-insights (Ours)	0.8132
2	Anti_UAV	0.8111
3	qiuye123	0.8000
4	wish	0.7609
5	feifei123	0.7438
6	number_one	0.7384
7	cxl	0.7239
8	hhh__	0.6805
9	jayeo	0.6695
10	yjwong1999	0.6630

Table 2. Ablation studies of the object detection.

Model	image-size	AP_{50}	AP_{50-95}
Baseline	640	84.5%	46.6%
+Shape-IoU	640	86.1%	47.9%
+NWD	640	86.6%	47.1%
+Shape-NWD	640	86.3%	47.8%
+Wise-IoU	640	85.1%	46.8%
+SSQ	640	<u>87.8%</u>	<u>48.7%</u>
+SSQ	1280	93.9%	53.4%

Table 3. Tracking Accuracy of Different models

	Model	MOTA
1	Baseline	0.735
2	+CMC	0.770
3	+L2 Fusion	0.775

method achieves state-of-the-art performance with 87.8% AP_{50} and 48.7% AP_{50-95} , outperforming all existing IoU variants (Wise-IoU, Shape-IoU, NWD) by significant margins. This highlights SSQ’s superiority in balancing localization accuracy and robustness for small UAV targets. When scaling the input resolution to 1280 (a standard practice for small-object detection), our SSQ-enhanced model demonstrates substantial gains, achieving 93.9% AP_{50} and 53.4% AP_{50-95} . This performance leap confirms the effectiveness of our strategy to employ higher resolutions for improved small-target recognition. The superior results solidly establish SSQ as a pivotal element of the “Strong Detector” framework, directly contributing to more robust multi-object tracking performance for infrared drones in subsequent processing stages.

4.3.2. Object Tracking

In our experiments, we used an improved version of SORT with a confidence-based cost matrix for data association as

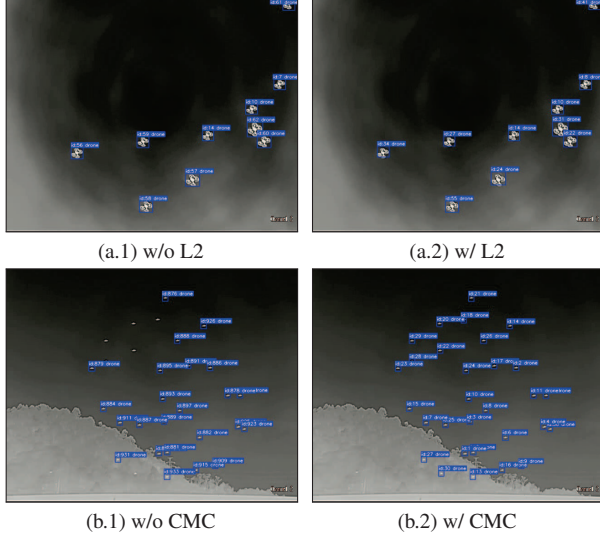


Figure 3. Comparison of tracking results w/o and w/ L2 or CMC on the same video sequence

the baseline tracking model. During experiments, we set the confidence score threshold to 0.01 and the new track score threshold to 0.25. To assess the individual contribution of each proposed module to tracking performance, we conducted ablation studies by incrementally introducing these enhancements to the baseline model. The experimental results are presented in Table 3. Introducing the CMC module to the baseline significantly improved tracking performance, increasing the MOTA from 0.735 to 0.770. This improvement clearly indicates that CMC reduces mismatches caused by changes in camera viewpoint. Furthermore, integrating the L2-IoU-based fusion strategy into the data association step further improved tracking accuracy, increasing the MOTA to 0.775. This incremental improvement demonstrates that the L2-IoU cost calculation method provides a more accurate representation of the spatial relationships between objects, reducing false negatives and ID switches during data matching, as shown in Figure 3. The ablation results clearly highlight the effectiveness of each introduced component, confirming that both the CMC module and the L2-IoU fusion strategy significantly enhance the accuracy of our tracking model.

5. Conclusion

In this paper, we propose Dist-Tracker, a tracking-by-detection framework specifically designed to address challenges in UAV multi-object tracking tasks. Our method integrates a SSQ loss based detector built upon YOLOv12 to improve small object detection robustness, alongside an enhanced SORT (FLIT) using L2-IoU Fusion for precise data association and CMC for camera movements. Experiments

conducted on the dataset from the 4-th Anti-UAV Challenge Track 3 demonstrated that our Dist-Tracker achieves the best performance in terms of MOTA. In future work, we will further improve the performance of the detector and introduce additional methods to enhance the accuracy of data association.

6. Acknowledgment

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